Sample Point: The outcome of a random experiment is called a sample point, which is an element in S.

Example 8.1

Consider the random experiment of tossing a coin once "Head" and "Tail" are the two possible outcomes. The sample space is S = {H, T}. It is a finite sample space which is presented in fig. 8.1.



Fig. 8.1. Tossing of a Coin Once

Example 8.2

Suppose that a study is conducted on all families with one or two children. The possible outcomes, in the order of births, are: boy only, girl only, boy and girl, girl and boy, both are boys and both are girls. Then, the sample space is S = {b,g,bg,gb,bb,gg}. It is also a finite sample space. Here, 'b' represents the child is a boy and 'g' represents the child is a girl.

Example 8.3

Consider the experiment of tossing a coin until head appears. Then, the sample space of this experiment is S = { (H), (T,H), (T,T,H), (T,T,T,H), … }. This is a countable sample space. If head appears in the first trial itself, then the element of S is (H); if head appears in the second attempt then the element of S is (T,H); if head appears in the third attempt then the element of S is (T,T,H) and so on.

Example 8.4

In the experiment of observing the lifetime of any animate or inanimate things, the sample space is

$$S = \{x: x \geq 0\},$$

where x denotes the lifetime. It is an example for uncountable sample space.